E D I P T

DESIGN THINKING

COURSE SLIDES · WSQ

Design Thinking Course for Businesses

WSQ Course Code: TGS-2026064719

TSC: Design Thinking Practices · DSN-ACE-3014-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v15.0 · 10 August 2026

COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

1

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code generated from the SSG portal.

2

Scan with your phone

Open your mobile phone camera and scan the QR code shown on screen.

3

Submit — three times

Attendance is taken at AM, PM and again before the Assessment.

⚠ 75% minimum attendance

Mandatory for all WSQ-funded courses. You must meet the 75% attendance rate based on the SSG Digital Attendance record to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS

PhD; ACTA/ACLP-certified adult educator with design thinking and innovation facilitation experience.

DELIVERS

WSQ courses on design thinking, innovation, digital transformation and emerging technology.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- A product or service experience you love — and one that frustrates you.
- What you want to be able to design or improve after this course.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

Download Your Course Material

lms-tms.tertiaryinfotech.com

My Courses › Design Thinking for Businesses

- **Trainer Slides** — ↓ PPT
- **Learner Slides** — ↓ PDF
- **Learner Guide** — ↓ PDF
- **Lesson Plan** — ↓ PDF
- **Activities / Labs** — ↓ FOLDER

1. **Sign in**
   Use the e-mail you registered with
2. **Open My Courses**
   Select this course from your list
3. **Click ↓ Download**
   Slides, Learner Guide and Lesson Plan
4. **Keep them open**
   Your assessment is open book

BROWSER-BASED · NO INSTALL REQUIRED

Your Lab Ed-Tools

1	**Design Thinking Toolkit** alfredang.github.io/designthinking/	Empathy Map, Persona, Journey Map, POV, HMW, Brainstorm, Prototype and Test canvases — the primary tool for this course.
2	**RACI Matrix** alfredang.github.io/raci/	Assign Responsible / Accountable / Consulted / Informed roles so a design idea has a clear owner when it moves to delivery.
3	**Scrum Planner** alfredang.github.io/scrum/	Turn validated design concepts into a prioritised product backlog and sprint plan.
4	**Digital Transformation Canvas** alfredang.github.io/digitaltransformation/	Position a design initiative within the organisation's wider transformation roadmap.
5	**Business Continuity Management** alfredang.github.io/bcm/	Stress-test a design solution for operational risk and continuity before rollout.

Skills Framework Alignment

1

TSC Title

Design Thinking Practices

2

TSC Code

DSN-ACE-3014-1.1

3

Abilities assessed

A1–A6: apply methodologies, uncover opportunities, use metrics, embed practice, facilitate prototypes, communicate outcomes.

4

Knowledge assessed

K1–K7: concepts, importance, traits, use cases, organisational approaches, methods and metrics.

SCHEDULE

Lesson Plan — 1 Day, 8 Course Hours

Morning

- Morning — 9:30am to 12:25pm
  - Digital Attendance (AM), introductions, learning outcomes
  - Topic 1: Key Concepts & Principles (Activities 1–2)
  - Topic 2: Applications of Design Thinking (Activity 3)
  - Tea break taken within training time
  - Lunch break 12:25pm – 1:25pm

Afternoon

- Afternoon — 1:25pm to 6:30pm
  - Digital Attendance (PM)
  - Topic 3: Action Phases — the Wallet Project (Activity 4)
  - Topic 4: Methodologies & Visual Tools (Activities 5–7)
  - Tea break taken within training time
  - Course feedback, TRAQOM survey and Final Assessment

WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Understand

Key concepts of design thinking, to communicate design outcomes.

2

LO2 · Apply

Design thinking to generate new ideas for the organisation.

3

LO3 · Uncover

Opportunities for applying design thinking.

4

LO4 · Implement

Plans to embed the stages of design thinking across the organisation.

5

LO5 · Execute

The design concept through prototypes.

6

LO6 · Measure

Outcomes of design ideas and solutions using metrics.

BEFORE YOU SIT THE ASSESSMENT

Briefing for Assessment

1. **Phones away**
   Place phones and other materials under the table or on the floor.

2. **No photos**
   No photographs or recording of the assessment scripts.

3. **No discussion**
   Work individually — no discussion during the assessment.

4. **Black or blue pen**
   Use a black or blue pen for hard-copy assessments.

5. **No correction fluid**
   No liquid paper or correction tape may be used.

6. **Time is up**
   Scripts are collected when the time is up.

FINAL ASSESSMENT

Assessment

Practical Performance (PP)

Design-thinking tasks

70 minutes · Open book

Tests your ABILITY to apply the methods using the course ed-tools.

Oral Questioning (OQ)

Verbal questions from the assessor

20 minutes · Open book

Tests your KNOWLEDGE of design thinking concepts, phases, tools and metrics.

📖 **Open book · you must be assessed 'Competent' in both instruments**

Permitted materials: the course slides, the Learner Guide and approved materials only. A minimum of 75% attendance is required to be eligible. An appeal process is available if required.

ON ASSESSMENT DAY

Assessment Flow

1
TRAQOM
Scan the QR code on the LMS

2
Attendance
Scan the SSG QR code

3
Assessment
Sit the PP, then answer the OQ verbally

4
Submit
Submit your PP answers on the LMS

5
Sign off
Sign the Assessment Summary Record

ELIGIBILITY

Criteria for Funding

- A minimum attendance rate of 75%, based on the SSG Digital Attendance record.
- Complete the assessment and be assessed as 'Competent'.
- Complete the mandatory TRAQOM survey on the LMS/TMS portal.

01

TOPIC 01

Key Concepts & Principles of Design Thinking

Concept · Importance · Traits of a design thinker · Applying design thinking for the organisation

SKILLS MAPPED: A6, K1, K2, K3, A1, K6

Key Concepts — Key Concepts & Principles of Design Thinking

1

What design thinking is

A human-centred problem-solving approach that aims to improve people's experience — as much a mindset as a process.

2

Why it matters

Design-led organisations report higher market share, stronger competitive advantage and more loyal customers.

3

Not just for designers

Anyone can think like a designer to create products and services that improve the customer experience.

4

Analytical vs creative

Design thinking deliberately pairs left-brain analysis with right-brain creativity.

5

The design thinker's mindset

Think users first, ask the right questions, sketch, commit to explore, prototype to test.

6

Desirability · Viability · Feasibility

Organisational design thinking lives at the intersection of user desire, business viability and technical feasibility.

“

You've got to start with the customer experience and work backwards to the technology.

Steve Jobs

Co-founder, Apple Inc.

TOPIC 01 · CONCEPT

What is Design Thinking?

1

A problem-solving approach

Aims to improve people's experience, not just ship a product.

2

Human-centred

Starts from a deep understanding of customers' needs and wants.

3

Creative and wide

Encourages consideration of a wide array of innovative solutions.

4

A mindset, not only a process

How you think about the problem matters as much as the steps you follow.

> **Design thinking is a human-centered approach to innovation that integrates the needs of people, the possibilities of technology, and the requirements for business success.**

Tim Brown

British industrial designer · President & CEO of IDEO

WHY IT MATTERS

Benefits of Design Thinking

1. Offers a problem-solving roadmap
2. Helps unlock innovative ideas
3. Solves real human problems
4. Creates behavioural change
5. Increases customer satisfaction
6. Gains competitive advantage

DESIGN-LED ORGANISATIONS OUTPERFORM

The ROI of Design Thinking

41%	46%	50%	70%
higher market share	competitive advantage overall	more loyal customers	digital experiences beat competitors

Source: Design Management Institute — dmi.org/page/2015DVIandOTW

DESIGN THINKING USES BOTH

Analytical and Creative Thinking

Analysis

- Left brain
  - Analytical
  - Rational
  - Objective
  - Present & past
  - Facts
  - Order and pattern
  - Planned

Creativity

- Right brain
  - Creative
  - Holistic
  - Subjective
  - Present & future
  - Feelings
  - Spatial
  - Spontaneous

Mindset — Traditional vs Design Thinker

Business-centric

- Traditional thinker

– "We have this problem — let's get in a room and brainstorm solutions."

– "Our competitors just launched X; how can we do X quickly?"

– "We have this technology — what can we use it for?"

– Starts from the business or the technology

Customer-centric

- Design thinker

– Think users first

– Ask the right questions

– Believe you can sketch

– Commit to explore

– Prototype to test and evaluate

– Starts from the human being served

MINDSET

Design thinking is a growth mindset.

Talent and ability are a starting point, not a ceiling — the same belief that lets a team keep iterating instead of defending its first idea.

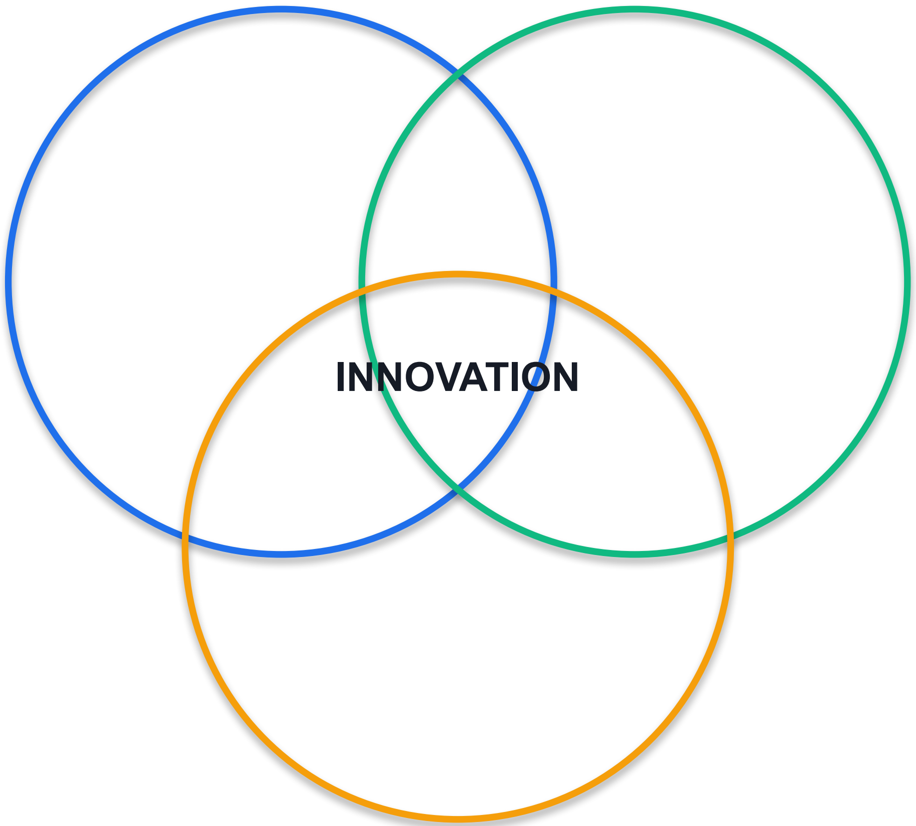
THE THREE LENSES

Design Thinking for the Organisation

DESIRABILITY
Do our customers want it?

VIABILITY
Does it work for the business?

FEASIBILITY
Can technology deliver it?



TOPIC 01 · HANDS-ON

Design a Vase — Reframing the Brief

ACTIVITY 1

A two-round warm-up that shows the difference between designing an object and designing an experience. In round 1 you design a vase. In round 2 you re-design against a human-centred brief and compare how much wider the solution space becomes.

You'll produce

Two sketched concepts and a written reflection on how the framing of a brief drives creativity.

Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), draw.chat, Padlet

ACTIVITY 1 · CHECK YOUR WORK

Design a Vase — Reframing the Brief

✅ You're done when…

You have two visibly different concepts on your board, and you can state in one sentence why the second brief produced a wider range of ideas than the first.

TOPIC 01 · HANDS-ON

Design Thinking Readiness — Mindset & Culture Self-Assessment

ACTIVITY 2

Benchmark your own team against the traits of a design thinker and the three lenses of organisational design thinking — desirability, viability and feasibility — then identify the single biggest barrier to embedding design thinking where you work.

You'll produce

A completed readiness scorecard with a prioritised list of cultural barriers and first actions.

Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), Digital Transformation Canvas (alfredang.github.io/digitaltransformation)

ACTIVITY 2 · CHECK YOUR WORK

Design Thinking Readiness — Mindset & Culture Self-Assessment

✅ You're done when…

Your scorecard is complete across all three lenses, and you have named one barrier and one specific, time-bound action to address it.

TOPIC RECAP

Recap — Key Concepts & Principles of Design Thinking

- You can now: experience how a narrowly framed brief limits creativity, and how reframing around the user unlocks it
- You can now: assess your organisation's design-thinking mindset and identify what must change

Tea Break

15 minutes

02

TOPIC 02

Applications of Design Thinking

Use cases across industries · Uncovering opportunities for applying design thinking

SKILLS MAPPED: A2, K5

Key Concepts — Applications of Design Thinking

1 **Airbnb**
Built a culture of experimentation and design-led iteration to move from failing start-up to a billion-dollar business.

2 **IBM**
Applies design thinking at scale to complex teams, problems and organisations through IBM Design Thinking.

3 **Bank of America**
Partnered with IDEO to research real savers, producing the 'Keep the Change' account.

4 **Uber Eats**
Designs its food-delivery service around observed customer, courier and restaurant journeys.

5 **Healthcare**
Stanford's Hasso Plattner Institute of Design redesigned the emergency-room patient experience.

6 **Public & social sector**
Clean Team (Ghana in-home toilets), Golden Gate Regional Center and The Good Kitchen redesigned public services.

CORPORATE USE CASES

Design Thinking in Business

Airbnb

- Culture of experimentation
- Design-led iteration
- Failing start-up → billion-dollar business

IBM

- IBM Design Thinking at scale
- Complex teams and organisations
- Restructured how products are built

Bank of America

- Partnered with IDEO
- Researched real savers
- 'Keep the Change' account

SERVICE & PUBLIC SECTOR USE CASES

Design Thinking in Services & Society

Uber Eats

- Designs around observed journeys
- Customer, courier and restaurant
- Continuous field research

Healthcare

- Stanford Hasso Plattner Institute
- Redesigned the ER experience
- Reduced patient anxiety

Public & social

- Clean Team — in-home toilets, Ghana
- Golden Gate Regional Center
- The Good Kitchen, Denmark

WHERE TO START

Uncovering Opportunities in Your Organisation

1

Look for friction

Where do customers or staff visibly struggle, complain or work around the system?

2

Follow the complaints

Repeated complaints and support tickets are free user research.

3

Watch the workarounds

A workaround is a user telling you the design is wrong.

4

Find the drop-off

Where do people abandon the journey — and what happened just before?

5

Ask the frontline

The people serving customers already know where the pain is.

6

Prioritise

Rank opportunities by impact and effort before committing a team.

Uncover Design Thinking Opportunities in Your Business

ACTIVITY 3

Study how Airbnb, IBM, Bank of America, Uber Eats and the public-sector cases applied design thinking, extract the transferable pattern from each, then map three concrete opportunities in your own business and rank them by impact and effort.

You'll produce

An opportunity map with three ranked design-thinking opportunities for your own organisation.

Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), Digital Transformation Canvas (alfredang.github.io/digitaltransformation), Padlet

ACTIVITY 3 · CHECK YOUR WORK

Uncover Design Thinking Opportunities in Your Business

✅ You're done when…

You can name three design-thinking opportunities in your own organisation, each with an identified user group and observable symptom, and justify which one you would tackle first.

TOPIC RECAP

Recap — Applications of Design Thinking

- You can now: analyse published use cases and uncover opportunities for applying design thinking in your own organisation

Lunch Break

1 hour

03

TOPIC 03

Action Phases of Design Thinking

The Stanford d.school five-stage model · Empathize · Define · Ideate · Prototype · Test

SKILLS MAPPED: A4, K4, A1

Key Concepts — Action Phases of Design Thinking

1

Empathize

Understand the user's experience, situation and emotion by observing, engaging and immersing — without judging.

2

Define

Synthesise the findings into a meaningful, actionable problem statement (user + need + insight).

3

Ideate

Go wide: generate a large variety and quantity of ideas before narrowing to the strongest few.

4

Prototype

Build to think — a cheap, fast, low-fidelity artefact that makes the idea tangible.

5

Test

Put the prototype in users' hands, gather feedback, and refine or reframe.

6

Iterative, not linear

The five phases loop — insights from Test routinely send you back to Empathize or Define.

STANFORD d.school MODEL

The Five Action Phases

1

Empathize

Understand the experience of the user you are designing for.

2

Define

Synthesise findings into an actionable problem statement.

3

Ideate

Generate a wide variety of possible solutions.

4

Prototype

Build a tangible representation of your idea.

5

Test

Put it in users' hands and gather feedback.

Action Phase 1 — Empathize

1. **Objective**
   Understand the experience, situation and emotion of the user you are designing for.

2. **Observe**
   View users and their behaviour in the context of their lives. Don't judge.

3. **Engage**
   Interact with people in conversations and interviews. Ask why — repeatedly.

4. **Immerse**
   Experience what your user experiences, first-hand.

5. **Beginner's mindset**
   Leave your assumptions behind; question what you think you already know.

6. **Example — the MRI scanner**
   Immersion revealed that an MRI room terrifies a child. The redesign turned the scan into an adventure.

Action Phase 2 — Define

1
Objective
Process and synthesise the findings into a problem statement you will address.

2
User
Develop an understanding of the persona you are designing for.

3
Needs
Select a limited set of needs to fulfil — needs should be verbs.

4
Insights
Express the insights you developed and define your design principles.

5
A good problem statement
Is human-centred, broad enough for creativity, narrow enough to be manageable.

6
Methods
Clustering, empathy mapping, POV statements, 'How Might We', why-how laddering.

> A problem well-stated is a problem half-solved.

Charles F. Kettering

Inventor · Head of research, General Motors

Action Phase 3 — Ideate

1 **Objective**
Focus on idea generation — translate problems into solutions.

2 **Go wide**
Explore a large quantity and variety of ideas to get beyond the obvious.

3 **Creativity**
Combine rational thought with imagination; defer judgement.

4 **Group synergy**
Leverage the group to reach new ideas and build on others'.

5 **Diverge, then converge**
Separate the generation of ideas from their evaluation.

6 **Facilitate actively**
Create a curious, courageous and concentrated atmosphere.

“

If at first the idea is not absurd, then there is no hope for it.

Albert Einstein

Theoretical physicist

Action Phase 4 — Prototype

1 **Objective**
Build to think — a simple, cheap and fast way to shape ideas you can interact with.

2 **It can be anything**
A wall of post-its, a role-play, a space, an object, an interface, a storyboard.

3 **Low fidelity**
Basic models using simple materials — fast, cheap, easy to change.

4 **High fidelity**
Closer to the finished product — used later, when the concept is settled.

5 **Just start building**
Don't spend too much time; remember what you're testing for.

6 **Build with the user in mind**
The prototype exists to provoke a reaction, not to impress.

Action Phase 5 — Test

1
Objective
Ask for feedback, learn about your user, reframe your POV and refine the prototype.

2
Show, don't tell
Put it in their hands and let them use it. Avoid over-explaining.

3
Create experiences
Let people talk about how they experience it and how they feel.

4
Compare
Let users test multiple prototypes to reveal latent needs and preferences.

5
Test in context
Use the natural setting where the product would really be used.

6
Negative feedback is data
Difficulty reveals new problems — revisit your solutions and reframe.

“

Insanity is doing the same thing over and over again and expecting different results.

Albert Einstein

On why testing must actually change the design

ITERATE

The five phases are a loop, not a line.

Insights from Test routinely send you back to Empathize or Define. Iteration is the method, not a sign of failure.

The Wallet Project — A Full Design Thinking Cycle

ACTIVITY 4

The classic Stanford d.school immersive exercise. Working in pairs, you take a partner through all five action phases — Empathize, Define, Ideate, Prototype and Test — to redesign their wallet, or more precisely the experience of carrying what matters to them. It is deliberately fast so you feel the rhythm of the whole cycle.

You'll produce

A tested paper prototype of a redesigned wallet, plus your partner's feedback and a refined concept.

Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), draw.chat, Padlet

ACTIVITY 4 · CHECK YOUR WORK

The Wallet Project — A Full Design Thinking Cycle

✅ You're done when…

You have completed all five phases, your prototype changed as a direct result of your partner's test feedback, and you can state the POV insight in your partner's own words.

Recap — Action Phases of Design Thinking

- You can now: run all five action phases end to end in a single compressed cycle

Tea Break

15 minutes

04

TOPIC 04

Methodologies and Visual Tools for Design Thinking

Empathize · Define · Ideate · Prototype · Test methods, tools and metrics

SKILLS MAPPED: A5, K7, A3

Key Concepts — Methodologies and Visual Tools for Design Thinking

1

Empathize tools

Empathy Map (Says / Thinks / Does / Feels), Persona Mapping, Journey Mapping, interviews and bodystorming.

2

Define tools

Point of View (POV) statements, 'How Might We' questions, Why-How laddering and the Business Model Canvas.

3

Ideate tools

Brainstorming, mind mapping, sketchstorm, analogies and gamestorming; then dot voting or a Now-Wow-How matrix to select.

4

Prototype tools

Low- and high-fidelity prototypes: paper sketches, storyboards, wireframes, cardboard and clay models.

5

Test tools

User test scripts, feedback grids, evaluation matrices and A/B comparison of alternative prototypes.

6

Measuring outcomes

Traditional KPIs, customer feedback (CSAT/NPS), design-thinking activity metrics and business-value/novelty mapping.

Methods by Action Phase

1

Empathize

Empathy map · Persona map · Journey map · Interviews · Bodystorming

2

Define

POV statement · How Might We · Why-how laddering · Business Model Canvas

3

Ideate

Brainstorm · Mind map · Sketchstorm · Analogies · Gamestorming · Dot voting

4

Prototype

Sketches · Storyboards · Wireframes · Paper and 3D models

5

Test

User test scripts · Feedback grids · Evaluation matrix · A/B comparison

6

Measure

KPIs · CSAT and NPS · Activity metrics · Business value vs novelty

EMPATHIZE TOOL

The Empathy Map

SAYS

Direct, verbatim quotes from the interview. Don't paraphrase — the exact words carry the insight.

THINKS

What the user is thinking but not saying out loud. Worries, doubts, judgements.

USER

DOES

The observable actions: what they actually do, where, and how long it takes.

FEELS

The emotional state and its intensity: anxious, rushed, proud, relieved.

GUIDELINES

Reading an Empathy Map

1

Look for contradictions

A positive quote beside a negative feeling is where the real insight hides.

2

Users are complex

Juxtaposition between quadrants is normal — and extremely useful.

3

Quote, don't summarise

Verbatim language keeps the user's meaning intact.

4

Turn it into a persona

Distil the map into a named persona with goals, frustrations and a defining quote.

EMPATHIZE TOOLS

Persona & Journey Mapping

1

Persona mapping

Identify who your customers are and how they make decisions, then design for that specific person.

2

Journey mapping

Chart every touchpoint over time — before, during and after the core interaction.

3

Find the pain points

The journey map exposes where the experience breaks down.

4

Find the moments that matter

Not every touchpoint is equal — some disproportionately shape the whole experience.

DEFINE TOOL

Point of View — The POV Formula

1

USER

The specific person you are designing for — not 'everyone'.

2

NEED

What they need to be able to do. Always a verb, never a solution.

3

INSIGHT

The surprising discovery that explains the need.

DEFINE TOOL

POV Examples

1

Example 1

A busy working mom needs a way to bring bags with her when away from home, because she buys the equivalent of three dog-poop bags a day.

2

Example 2

A teacher needs a way to clean and store used sandwich bags at school, because he discards a new bag each morning rather than washing it.

3

The template

[USER] needs a way to [USER'S NEED] because [INSIGHT].

4

State your assumptions

Write down what must be true for the POV to hold — then test it.

IDEATE TOOLS

Ideation Methods

1

Generate

Brainstorm · Mind map · Sketchstorm · Storyboard · Analogies · Provocation

2

Go further

Bodystorm · Gamestorming · Cheatstorm · Crowdstorm · Co-creation workshops

3

Brainstorm rules

Set a mission · Limit the time · Defer judgement · Build on ideas · Go for volume

4

Select the best

Dot voting · Four categories · Bingo selection · Affinity maps · Now-Wow-How matrix

5

Remember

10 ideas beat 3. 200 ideas beat 50. Quantity breeds quality.

6

Mind mapping

Start from the central problem and branch outwards to see connections you'd otherwise miss.

PROTOTYPE TOOLS

Low- vs High-Fidelity Prototyping

Start here

- Low fidelity
  - Paper sketches and storyboards
  - Cardboard, foam, clay, building blocks
  - Fast, cheap, easy to throw away
  - Use early, to test the concept
  - Invites honest criticism — it clearly isn't finished

Later

- High fidelity
  - Wireframes and interactive mock-ups
  - Near-final materials and finishes
  - Slower and more expensive
  - Use later, to test the detail
  - Risk: people critique the polish, not the idea

TEST TOOLS

Testing Your Prototype

1 **Prototype as if you're right…**
…but test as if you know you're wrong.

2 **Find the cheapest test**
What is the fastest, cheapest test that could disprove your key assumption?

3 **Define success up front**
What metric will tell you it worked — pre-orders, votes, completion rate, smiles?

4 **Let users compare**
Offer alternatives so the user can express a preference, not just be polite.

5 **Ask them to think aloud**
Narration reveals confusion that observation alone will miss.

6 **Then revise**
Revise and advance the idea until it genuinely meets the need.

Measuring the Outcome of Design Ideas

1

Traditional KPIs

Increased sales, ROI per project and other financial measures.

2

Customer feedback

Customer satisfaction, Net Promoter Score, campaign response, usability metrics.

3

Design thinking activity

Number of projects run, people trained, coaches developed.

4

Quick results

Concepts finished, projects launched, funded or in development.

5

Beyond execution

Track creative behaviours, not only delivery throughput.

6

Three business drivers

Understand customers better · protect share from disruption · develop innovative team dynamics.

Measuring Business Value and Novelty

VALUE →

Valuable · Novel The target quadrant — genuinely new and worth doing. Protect and resource these.	**Valuable · Not novel** Solid incremental improvement. Reliable returns, little differentiation.
Not valuable · Novel Interesting but unprofitable. A nice demo, not a business case.	**Not valuable · Not novel** Stop. This consumes capacity without creating value.

NOVELTY →

Empathy Map & Persona — The Gift-Giving Experience

ACTIVITY 5

Redesign the gift-giving experience for a partner. You start by building empathy: interview them about the last gift they gave, then plot what they said, thought, did and felt into a four-quadrant empathy map and distil it into a persona.

You'll produce

A completed four-quadrant empathy map and a one-page persona for your partner.

Ed-tools: Design Thinking Toolkit — Empathy Map & Persona canvases (alfredang.github.io/designthinking)

ACTIVITY 5 · CHECK YOUR WORK

Empathy Map & Persona — The Gift-Giving Experience

✅ You're done when…

All four quadrants are populated with your partner's own words, you have identified at least one contradiction between quadrants, and your partner confirms the persona sounds like them.

POV, How Might We & Brainstorming

ACTIVITY 6

Take the empathy map from the previous lab and convert it into a sharp Point of View problem statement, reframe it as a set of 'How Might We' questions, then run a timed brainstorm and converge on the best ideas using dot voting.

You'll produce

A validated POV statement, three HMW questions, 20+ generated ideas and a shortlist of three.

Ed-tools: Design Thinking Toolkit — POV, HMW and Brainstorm canvases (alfredang.github.io/designthinking)

ACTIVITY 6 · CHECK YOUR WORK

POV, How Might We & Brainstorming

✅ You're done when…

Your POV has a verb-based need and a genuine insight, you generated 20 or more ideas, and you have three shortlisted ideas each with a stated reason for selection.

Prototype, Test & Measure the Outcome

ACTIVITY 7

Turn your shortlisted idea into a low-fidelity prototype, test it with a real user, then define how you would measure whether it succeeded — moving from 'we like this idea' to 'here is the evidence it works'. Finish by handing the concept over for delivery with a RACI matrix and a sprint plan.

You'll produce

A tested prototype, a completed evaluation matrix, an ROI/metrics plan and a RACI handover.

Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), RACI Matrix (alfredang.github.io/raci), Scrum Planner (alfredang.github.io/scrum), BCM (alfredang.github.io/bcm)

ACTIVITY 7 · CHECK YOUR WORK

Prototype, Test & Measure the Outcome

✅ You're done when…

Your prototype was changed by user feedback, you have named the assumption you tested and the metrics that would prove success, the concept has an owner in the RACI matrix and a first sprint goal, and you have pitched the outcome and its value to your stakeholder leading with the insight rather than the solution.

TOPIC RECAP

Recap — Methodologies and Visual Tools for Design Thinking

- You can now: build an empathy map and persona to capture what a user says, thinks, does and feels
- You can now: convert empathy findings into a POV problem statement, generate a high volume of ideas and select the strongest
- You can now: build a low-fidelity prototype, test it with a user, define the metrics that prove it worked, and communicate the outcome and its value to stakeholders

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES

What You Achieved

1

LO1 · Understood

The key concepts, principles and business value of design thinking.

2

LO2 · Applied

Design thinking methods to generate new ideas for your organisation.

3

LO3 · Uncovered

Real opportunities to apply design thinking where you work.

4

LO4 · Implemented

A plan to embed the five action phases across the organisation.

5

LO5 · Executed

A design concept as a tested low-fidelity prototype.

6

LO6 · Measured

The outcome of a design idea using metrics and a value/novelty map.

NEXT STEPS

Take It Back to Work

1. **Pick one opportunity**
   Start with the impact-versus-effort winner from Activity 3.

2. **Run a one-week sprint**
   Empathize and Define on days 1–2, Ideate day 3, Prototype and Test days 4–5.

3. **Give it an owner**
   Use the RACI matrix so the concept has an accountable name against it.

4. **Define the metric first**
   Agree what success looks like before you build anything.

5. **Keep the ed-tools**
   All the canvases stay free and available in your browser after the course.

6. **Share the result**
   Show colleagues the prototype and the user feedback — evidence beats opinion.

YOURS TO KEEP · FREE · NO LOGIN REQUIRED

Keep Using These Tools After the Course

1	**Design Thinking Toolkit** alfredang.github.io/designthinking/	Re-run the Empathy Map and POV canvases on a live problem with your own team.
2	**RACI Matrix** alfredang.github.io/raci/	Assign an owner for your first initiative so it survives the handover to delivery.
3	**Scrum Planner** alfredang.github.io/scrum/	Turn the concept you prototyped today into a backlog and a first sprint goal.
4	**Digital Transformation Canvas** alfredang.github.io/digitaltransformation/	Position the initiative inside your organisation's wider transformation roadmap.
5	**Business Continuity Management** alfredang.github.io/bcm/	Stress-test the solution for operational risk before you commit to rollout.

WRAP-UP

Summary & Q&A

- Design thinking is a human-centred, iterative approach to solving real problems.
- The five action phases — Empathize, Define, Ideate, Prototype, Test — loop rather than run once.
- Methods and visual tools make each phase concrete and repeatable.
- Metrics turn a good idea into a defensible business case.
- Questions?

FINAL ASSESSMENT

Assessment

Practical Performance (PP)

Design-thinking tasks

70 minutes · Open book

Tests your ABILITY to apply the methods using the course ed-tools.

Oral Questioning (OQ)

Verbal questions from the assessor

20 minutes · Open book

Tests your KNOWLEDGE of design thinking concepts, phases, tools and metrics.

📖 Open book · you must be assessed 'Competent' in both instruments

Permitted materials: the course slides, the Learner Guide and approved materials only. A minimum of 75% attendance is required to be eligible. An appeal process is available if required.

ON ASSESSMENT DAY

Assessment Flow

1 TRAQOM

Scan the QR code on the LMS

2 Attendance

Scan the SSG QR code

3 Assessment

Sit the PP, then answer the OQ verbally

4 Submit

Submit your PP answers on the LMS

5 Sign off

Sign the Assessment Summary Record

Digital Attendance (Mandatory)

1
Trainer shows the QR

The trainer or administrator displays the digital attendance QR code generated from the SSG portal.

2
Scan with your phone

Open your mobile phone camera and scan the QR code shown on screen.

3
Submit — three times

Attendance is taken at AM, PM and again before the Assessment.

⚠ 75% minimum attendance

Mandatory for all WSQ-funded courses. You must meet the 75% attendance rate based on the SSG Digital Attendance record to be eligible for assessment and funding.

CONTINUE YOUR LEARNING

Recommended Courses

1. WSQ – Agile Project Management with Scrum
2. WSQ – Digital Transformation Strategy for Business
3. WSQ – Business Innovation with Emerging Technology
4. WSQ – Data Visualisation for Business Decision Making
5. WSQ – Customer Experience and Service Design
6. WSQ – Business Continuity Management Essentials

Support

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4

💬 After the course

Free post-course consultation on the subject matter is available to every learner.

KEEP DESIGNING

Thank You!

You can now apply design thinking to uncover real user needs, generate ideas, prototype them and prove they work.